

Early development of intractable epilepsy in children

A prospective study

A.T. Berg, PhD; S. Shinnar, MD, PhD; S.R. Levy, MD; F.M. Testa, MD; S. Smith-Rapaport, MS; and B. Beckerman, MS

Article abstract—*Background:* Little is known about early prediction of intractable epilepsy (IE) in children. Such information could help guide the early use of new therapies in selected patients. *Methods:* Children with newly diagnosed epilepsy ($n = 613$) were prospectively identified from child neurology practices in Connecticut (1993–1997) and followed-up for the occurrence of IE (failure of >2 drugs, >1 seizure/month, over 18 months). Etiology and epilepsy syndromes were classified per International League Against Epilepsy guidelines. *Results:* The median follow-up is 4.8 years, and 599 (97.7%) have been followed for more than 18 months. Sixty children (10.0%) have met the criteria for IE, including 34.6% with cryptogenic/symptomatic generalized, 2.7% with idiopathic, 10.7% with other localization-related, and 8.2% with unclassified epilepsy ($p < 0.0001$). After multivariable adjustment for epilepsy syndrome, initial seizure frequency ($p < 0.0001$), focal EEG slowing ($p = 0.02$), and acute symptomatic or neonatal status epilepticus ($p = 0.001$) were associated with an increased risk of IE, and age at onset between 5 and 9 years was associated with a lowered risk ($p = 0.03$). The absolute number of seizures and unprovoked or febrile status epilepticus were not associated substantially with IE. *Conclusions:* Approximately 10% of children meet criteria for IE early in the course of their epilepsy. Cryptogenic/symptomatic generalized syndromes carry the highest risk and idiopathic syndromes the lowest. Half of IE occurs in children with nonidiopathic localization-related syndromes. Initial seizure frequency is highly predictive of IE. By contrast, absolute number of seizures and unprovoked or febrile status epilepticus are not.

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Prospective population-based and other types of studies of remission in epilepsy have dramatically changed our understanding of the natural history of epilepsy, the need for treatment, and the merits of eventually stopping medications.^{1–7} Although most individuals do well, some develop medically refractory (intractable) epilepsy. Studies of remission do not directly address development of intractability, primarily because not being in remission is not necessarily the same as being refractory to treatment. This is especially true early in the disorder, particularly in children,^{8,9} many of whom are not even treated.^{10,11} No prospective studies, and few other types of studies,^{12–14} explicitly address the early risk and predictors of intractable epilepsy.

Because many of the large-scale long-term remission studies were started, the approach to studying epilepsy syndromes has become increasingly accept-

ed.¹⁵ Because this approach is relatively new, many existing prognostic studies focused on components of syndromes (e.g., age at onset, cause, seizure type) and not syndromes themselves. We currently do not know to what extent the syndrome explains many of the associations with outcome (usually remission) previously shown to be important. No studies examine intractability with respect to traditionally studied prognostic factors for remission simultaneously with the epilepsy syndrome to determine the relative importance of the syndrome versus factors that are components of syndromes.

Early prediction of intractability would be valuable, because rapid identification of patients at highest risk would allow physicians to consider them earlier for treatment with recently approved medications and other surgical and nonsurgical treatments. The following study was designed explicitly to deter-

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From Northern Illinois University, Department of Biological Sciences (Dr. Berg), DeKalb, IL; Montefiore Medical Center, Albert Einstein College of Medicine, Departments of Neurology, and Pediatrics, and the Comprehensive Epilepsy Management Center (Dr. Shinnar), Bronx, NY; and Yale University, Departments of Pediatrics (Drs. Levy, Testa, Smith-Rapaport, and Beckerman) and Neurology (Drs. Levy and Testa), New Haven, CT.

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Address correspondence and reprint requests to Dr. Anne T. Berg, Department of Biological Sciences, Northern Illinois University, DeKalb, IL 60115; e-mail: atberg@niu.edu

mine the risk of intractable epilepsy in children prospectively followed from their initial diagnosis of epilepsy and to examine the relative importance of traditional predictors of outcome in epilepsy and other factors in the context of the epilepsy syndrome.

Methods. Children were prospectively identified when first diagnosed with epilepsy by participating physicians (including 16 of the 17 child neurologists and selected pediatricians and adult neurologists) in Connecticut, January 1993 through December 1997. Active surveillance was used in each practice to identify all eligible children. Although our study is not strictly population-based, we recruited from academic centers, private practices, and community clinics to achieve a group representative of children with epilepsy in the state. Children had to experience the first unprovoked seizure between 1 month and 15 years of age, have at least two unprovoked seizures on separate days,¹⁶ and be newly diagnosed during the recruitment period by a participating physician. Written informed parental consent and written or oral assent of the child (as appropriate) were obtained. The Institutional Review Boards of all participating institutions approved all procedures.

Trained staff members interviewed the parents. Investigators reviewed all pertinent medical records. Methodologic details were presented previously.¹⁷ Study personnel conducted follow-up calls every 3 months. Medical records were reexamined at 6-month intervals.

We defined intractability as failure, for lack of seizure control, of more than 2 first-line anti-epileptic drugs (AED)¹⁸ with an average of more than 1 seizure per month for 18 months and more than 3 consecutive months seizure-free during that interval. Drugs had to be pushed to maximum tolerated doses and noncompliance ruled out as determined through medical record review and follow-up calls with the parents. Initial AED use in this cohort has been presented¹¹ and reflects current accepted guidelines for practice.¹⁸ An inappropriate drug was rarely given (e.g., carbamazepine for absence epilepsy). We did not consider that to be a drug failure for the purpose of defining intractability. Children who discontinued two AEDs for adverse reactions and failed one other for seizure control also satisfied the medication criterion. We focused on the early occurrence of intractability with onset during the first 2 years. Four children who met criteria only after 4 or more years are not considered intractable for the purposes of this report.

Epilepsy syndromes and seizures were classified based on all information available at diagnosis and according to International League Against Epilepsy's (ILAE) guidelines^{15,19} by three pediatric epilepsy specialists (S.S., S.L., F.T.).¹⁷ The syndromic classification recognizes some syndromes that have been well characterized and studied with respect to prognosis. These include the localization-related and generalized idiopathic syndromes (e.g., benign rolandic and childhood absence epilepsy), generally considered to have an excellent prognosis and the cryptogenic and symptomatic generalized syndromes, generally recognized as having a poor prognosis.²⁰ In addition, there are less well-differentiated syndromes. The symptomatic and cryptogenic localization-related epilepsies represent a broad spectrum of syndromes defined by cause and localization,

to the extent that these are known.¹⁵ There is also a small group with both clearly focal and generalized signs (e.g., Landau-Kleffner). The outcome in this group is mixed. The last category, "undetermined and without clear focal or generalized signs," is essentially for unclassifiable epilepsies and includes children who have generalized tonic-clonic seizures and normal EEGs.

Etiology¹⁶ was classified separately from syndrome although it partially depends on the syndrome. Remote symptomatic refers to the presence of an underlying neurologic condition or insult associated with an increased risk of epilepsy (e.g., history of bacterial meningitis, stroke, cerebral palsy).¹⁶ Idiopathic syndromes are almost always assigned an idiopathic etiology. Occasionally, a neurologic abnormality coexists with an idiopathic syndrome (e.g., childhood absence epilepsy with mental retardation), in which case we classified the etiology as remote symptomatic despite the idiopathic syndrome. Results of neuroimaging studies were used in classifying etiology. This has been fully described previously.²¹ Cryptogenic etiology refers to epilepsy that does not meet the criteria for an idiopathic syndrome and for which there is no identified significant underlying neurologic abnormality or condition. Such individuals appear to be otherwise neurologically normal.

Initial seizure frequency was defined as the ratio of the total number of unprovoked seizures since epilepsy diagnosis divided by the time between the dates of first unprovoked seizure and formal diagnosis. Every effort was made to assess the number of seizures as accurately as possible. Reasonable estimates were made when necessary (e.g., 5–7/week for 10 weeks was considered 6/week for a total of 60).

Chi-square and *t*-tests were used for bivariate comparisons. Log transformations were employed when necessary to normalize a highly skewed distribution. For some continuous variables, categories were constructed to facilitate presentation of data and testing of the assumption of linearity.

Multivariable analysis with the proportional hazards model permitted assessment of the relative importance of several factors simultaneously. The technique provides a rate ratio (RR), the ratio of the rate at which a group with a characteristic of interest develops an outcome relative to the rate of a reference group without the characteristic. An RR of 1.0 indicates no association; <1.0, a protective association; and >1.0, an increased risk associated with the factor. Statistical interactions were considered, as needed, during modeling procedures.

A separate analysis examined predictors of intractability just in children with nonidiopathic, localization-related epilepsy.

Results. A total of 613 children were recruited into the study. The median age at onset was 5.3 years, and half were boys. The median follow-up is 5.0 years. Of the original 613, 599 (97.7%) have been followed for longer than 18 months, the minimum time needed to meet the study criteria for intractability, and hence are included in this analysis. Overall, 60, 10.0% of the total and 13.3% of those with nonidiopathic syndromes, met the criteria for intractability with onset during the first 2 years after initial diagnosis of epilepsy (figure).

Bivariate analysis—full group. Syndromic grouping ($p < 0.001$) and etiology ($p < 0.001$) were strongly correlated

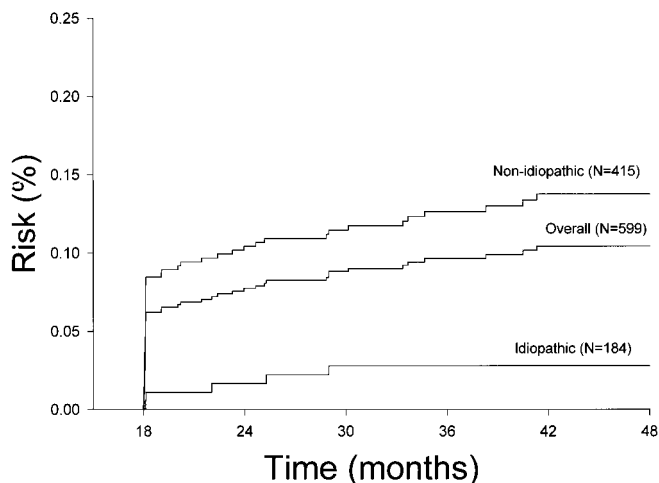


Figure. Risk of intractability as a function of time since diagnosis.

with the risk of intractability (table 1). The risk was highest in the cryptogenic/symptomatic generalized groups, 18 of 52 (34.6%); lowest in the idiopathic groups (including benign Rolandic and idiopathic generalized epilepsies), 5 of 184 (2.7%); and intermediate in the other localization-related (cryptogenic and symptomatic), 31 of 290 (10.7%), and undetermined (largely unclassified) groups, 6 of 73 (8.2%).

We also examined age, sex, abnormal neuroimaging, history of neonatal seizures and febrile seizures, unprovoked, febrile, and "other" (i.e., neonatal or other acute provoked) status epilepticus, multiple seizures in a single day, seizure types, and selected EEG characteristics (table 1). Age at onset and log transformation of number of seizures, duration of epilepsy, and initial seizure frequency/month, all before diagnosis, were examined. Strong, significant bivariate associations were seen for all of these except number of seizures (table 2).

Multivariable analysis, full group. On multivariable analysis, cryptogenic/symptomatic generalized and idiopathic syndromes, log initial seizure frequency, "other" (neonatal and acute provoked) status epilepticus, focal slowing, and onset age 5 to 9 years remained in the model (table 3).

There was unexpected interaction between cryptogenic/symptomatic generalized syndromes, seizure frequency, and myoclonic seizures. In the cryptogenic/symptomatic generalized syndromes, a high initial seizure frequency was protective against intractability in patients with myoclonic seizures. The small numbers with specific syndromes within this group precluded determining whether this was specific to myoclonic seizures per se or due to selected syndromes. This finding necessitated the inclusion of an interaction term for myoclonic seizures and seizure frequency within the cryptogenic/symptomatic generalized group. Myoclonic seizures remained in the model as a main effect for the interaction term, even though its main effect was not significant.²² Detailed examination of intractability within this heterogeneous group is beyond the scope of this report.

Age of onset less than 1 year as well as febrile and unprovoked status did not approach significance. The rate ratio for the log number of seizures was less than 1.0,

indicating a modestly protective effect. Duration of epilepsy no longer mattered. Almost all of the associations with age younger than 1 year, etiology, and abnormal neuroimaging seen in the bivariate analysis were explained by syndromic grouping in the multivariable analysis. The effect of having multiple seizures in a single day was entirely explained by initial seizure frequency.

Intractability in localization-related epilepsy. On bivariate analysis, several factors were associated with intractability in children with nonidiopathic localization-related syndromes, those in whom resective surgical interventions might be used if the epilepsy became intractable (tables 4 and 5). On multivariable analysis, the predominant predictor of intractability was log of initial seizure frequency (table 6). Focal slowing, neonatal seizures and "other" (neonatal and acute provoked) status epilepticus were statistically significant. Etiology, febrile seizures, febrile and unprovoked status, multiple seizures in a single day, and indicators for young age at onset did not approach significance. There was no association with the number of seizures before initial diagnosis.

To explore initial seizure frequency further, we examined just children with nonidiopathic localization-related epilepsy who presented with only two seizures ($n = 115$). Children with multiple seizures in a single day were excluded. The interval between the two seizures (essentially the inverse of seizure frequency in this case) was 0.5 months in those who met criteria for intractability and 4.7 months in those who did not ($p = 0.0001$).

Average follow-up since meeting criteria for intractability is 38 months. Only seven children have since experienced a 1-year remission 1.5 to 3.5 years after becoming intractable. Five eventually responded to topiramate ($n = 2$), lamotrigine ($n = 1$), vigabatrin ($n = 1$), or the ketogenic diet ($n = 1$). Another failed seven drugs and then responded to adrenocorticotrophic hormone (ACTH). The last had idiopathic generalized epilepsy and ultimately came under control. Ten children have died, three of underlying neurodegenerative conditions, six of other medical complications related to their etiology or to uncontrolled seizures, and one of reasons unrelated to epilepsy.

Discussion. Identification of children at high risk of early intractability permits planning for early intervention. Available remission studies do not provide the specific information needed to do this, and little is known about the risk and predictors of intractability early in the course of pediatric epilepsy. Overall, 10% of our cohort met the criteria for intractability (13.0% of those with nonidiopathic syndromes) early in the course of their disorder. The three most salient positive findings are the overwhelming prognostic significance of syndromic groupings, seizure frequency, and focal slowing on the EEG.

The finding for the cryptogenic/symptomatic generalized epilepsy syndromes is not surprising. These syndromes are notorious for their poor outcomes.²⁰ The absolute and relative magnitude of their effects has not been previously demonstrated in a large, community-based study. Also, over half of the intractability observed did not involve these syndromes.

Syndromic grouping and other factors also ex-

Table 1 Associations between intractability and categorical prognostic factors in children with >18 months' follow-up (n = 599)

Prognostic factor	No. followed >18 mo	No. (%) intractable	p Value (df if >1)
Sex			
Boys	302	28 (9.3)	0.54
Girls	297	32 (10.8)	
Age at onset, y			
<1	65	19 (29.2)	<0.001 (df = 4)
1–2	81	12 (14.8)	
3–4	92	9 (9.8)	
5–9	232	8 (3.5)	
10–15	129	12 (9.3)	
Syndrome			
Localization-related			
Idiopathic	59	1 (1.7)	<0.001 (df = 7)
Symptomatic	190	24 (12.6)	
Cryptogenic	100	7 (7.0)	
Generalized			
Idiopathic	125	4 (3.2)	<0.001 (df = 2)
Cryptogenic/symptomatic	43	13 (30.2)	
Symptomatic	9	5 (55.6)	
Undetermined with both focal and generalized features	5	0 (0)	
Undetermined without clear focal or generalized features	68	6 (8.8)	
Etiology			
Idiopathic*	182	5 (2.8)	<0.001 (df = 2)
Cryptogenic	309	32 (10.4)	
Remote symptomatic	108	23 (21.3)	
Neuroimaging (n = 477 with studies)			
Normal/incidental/equivocal†	418	42 (10.1)	0.03
Abnormal and etiologically relevant	59	12 (20.3)	
EEG‡			
Normal/equivocal	163	12 (7.4)	0.17 0.86 0.005 0.07
Abnormal	436	48 (11.0)	
Focal spikes	293	30 (10.2)	
Any slowing	106	19 (17.9)	
Focal slowing	58	10 (17.2)	
Seizure types§ (selected)			
Any generalized onset	272	23 (8.5)	0.24 0.004
Generalized tonic-clonic	130	5 (3.9)	
Myoclonic	52	11 (21.2)	0.012 0.27
Absence (typical or atypical)	110	8 (7.3)	
Atonic	10	3 (30.0)	0.08 0.28
Any partial onset	330	37 (11.2)	
Seizure of undetermined onset	18	4 (22.2)	0.12 0.91
Multiple seizure types	113	11 (9.7)	
Status epilepticus			
None‡	546	50 (9.2)	0.04 0.60 0.64 <0.001
Unprovoked	43	5 (11.6)	
Febrile only	7	1 (14.3)	
Other	5	5 (100)	
Multiple seizures in one day			
No	325	22 (6.8)	0.004
Yes	274	38 (13.9)	
Seizure frequency per mo			
<1	99	1 (1.0)	<0.001 (5 df)
1–	266	18 (6.8)	
10–	113	17 (15.0)	
50–	43	6 (14.0)	
100–	37	7 (18.9)	
200–	41	11 (26.8)	
Neonatal seizures§			
No	579	53 (9.2)	<0.001
Yes	16	7 (43.8)	
Febrile seizures§			
No	521	55 (10.6)	0.29
Yes	74	5 (6.8)	

* Two children with idiopathic syndromes also met criteria for remote symptomatic etiology.

† Incidental includes pineal, choroid fissure, and arachnoid cysts.

‡ Indicates reference group for χ^2 . For status, one child had both febrile and unprovoked status and met criteria for intractability. For seizure type and EEG, reference is to all others without specific characteristic.

§ History unavailable for four.

Table 2 Associations between intractability and continuous factors in the full group (n = 599)

Prognostic factor	Not intractable (n = 539), mean (SD)	Intractable (n = 60), mean (SD)	p Value
Age at onset, y	6.0 (3.9)	4.2 (4.4)	0.001
Log seizure frequency	1.5 (2.2)	3.2 (2.0)	<0.0001
Log duration of epilepsy, mo	1.1 (1.6)	0.0 (1.8)	<0.0001
Log number of seizures	2.6 (2.2)	3.3 (2.2)	0.04

plained almost all of the effect of symptomatic etiology, one of the more important predictors of remission and other outcomes in the literature.^{1, 3-5,12} In the earlier literature, those with idiopathic and cryptogenic etiology were grouped together. With the idiopathic cases now classified separately, the lack of substantial difference between those with a cryptogenic versus a remote symptomatic etiology could possibly reflect that fact that some of those in the cryptogenic group may indeed have underlying abnormalities that simply have not yet been appreciated. Once they are, such cases would then be reclassified as remote symptomatic.

Most studies began before current guidelines for classifying epilepsy syndromes¹⁵ and consequently focused on factors that are components of syndromes (e.g., age, etiology, seizure type) but that, individually, are not usually specific to any one syndrome.

Table 3 Multivariable model for predicting intractability in the full cohort followed ≥ 18 months (n = 599)

Prognostic factor	Rate ratio	95% Confidence interval	p Value
Cryptogenic/symptomatic generalized syndrome	3.01	1.32, 6.85	0.009
Idiopathic syndrome	0.23	0.09, 0.63	0.004
Log seizure frequency	1.41	1.23, 1.62	<0.0001
Focal slowing	2.31	1.13, 4.74	0.02
Age 5 to 9	0.42	0.19, 0.92	0.03
"Other" status epilepticus	5.96	2.00, 17.71	0.001
Myoclonic seizures*	1.21	0.38, 3.85	0.75
Interaction of myoclonic sz* log seizure frequency	0.73	0.57, 0.95	0.02
Etiology†	1.60	0.86, 2.98	0.14
Febrile status†	1.18	0.16, 8.61	0.87
Unprovoked status†	1.81	0.71, 4.65	0.22
Neonatal seizures†	1.94	0.75, 4.99	0.17
Log number of seizures†	0.86	0.72, 1.03	0.10
>1 Seizure in a day†	0.56	0.26, 1.21	0.14
Age <1 y at onset†	1.50	0.77, 2.91	0.23

* Main effect is included because of significant interaction within cryptogenic/symptomatic generalized group.

† Estimate is after adjustment for first eight terms and shows the residual effect of each factor after adjustment for the significant predictors. The first eight terms are not adjusted for non-significant terms.

Age and etiology, individually or together, were highly predictive of intractability in our overall cohort. The syndromic groupings, however, explained most of their predictive value.

The epidemiologic study of epilepsy syndromes has become a well-established endeavor.^{5,23-26} The classification provides a valuable framework for understanding prognosis. There are probably differences in the application of the classification by different investigators. For the classification to be truly useful, a high degree of uniformity and consistency in its use must be achieved across research groups and clinicians.

After syndromes, initial seizure frequency was the single strongest predictor. Few data address the prognostic significance of seizure frequency. A recent retrospective case-control study found a similar effect for <1 versus >1 seizure/month but did not control for syndrome.¹⁴ Seizure frequency (<1 versus >1 seizure/month) has been examined in studies of remission⁴ and relapse after stopping medications.²⁷ Our more detailed approach found an effect over a continuum of seizure frequency and even in those with only two seizures. This result was not explained by other factors. Few other data address this point. We note that two studies examined multiple seizures in a single day and found no influence on recurrence after an initial episode of unprovoked seizure.^{28,29} Conversely, in children with two seizures, an interval <6 months between the seizures was predictive of having many (>10) seizures.³⁰

The finding for seizure frequency contrasts with the lack of any increased risk associated with number of seizures. This challenges the theories of earlier investigators, who believed that seizures caused increasing degrees of damage to the brain and led to intractability.³¹ Most studies find the initial seizure number does not predict remission^{4,8,32,33} or relapse after stopping medications.^{34,35} Having more seizures before diagnosis probably does not improve outcome; however, the upper confidence limit rules out a substantial deleterious effect of seizure number per se.

Neonatal seizures and status epilepticus during the neonatal period or accompanying an acute insult were rare; however, when either occurred, the outcome was especially poor. Little is known about the prognostic importance of focal slowing per se.

Negative findings of interest include the lack of association between intractability and either unprovoked or febrile status, both overall and in those

Table 4 Associations between intractability and categorical prognostic factors in the group with nonidiopathic localization-related syndromes (*n* = 290)

Prognostic factor	No. followed > 18 mo	Intractable, n (%)	<i>p</i> Value (<i>df</i> if >1)
Sex			
Boys	137	15 (11.0)	0.89
Girls	153	16 (10.5)	
Age at onset, y			
<1	23	7 (30.4)	0.02 (4 <i>df</i>)
1–2	47	5 (10.6)	
3–4	48	4 (8.3)	
5–9	100	5 (5.0)	
10–15	72	10 (13.9)	
Syndrome grouping			
Symptomatic localization-related	190	24 (12.6)	0.13
Cryptogenic localization-related	100	7 (7.0)	
Etiology			
Cryptogenic	222	21 (9.5)	0.24
Remote symptomatic	68	10 (14.7)	
Neuroimaging (<i>n</i> = 256 with studies)			
Normal/incidental/equivocal	216	25 (11.6)	0.77
Abnormal and etiologically relevant	40	4 (10.0)	
EEG*			
Normal/equivocal	103	6 (5.8)	0.04 0.09 0.03 0.02
Abnormal	187	25 (13.4)	
Focal spikes	165	22 (13.3)	
Any slowing	64	12 (18.8)	
Focal slowing	46	10 (21.7)	
Seizure frequency per mo			
<1	59	1 (1.7)	0.002 (5 <i>df</i>)
1–	156	13 (8.3)	
10–	46	11 (23.9)	
50–	9	1 (11.1)	
100–	10	2 (20.0)	
200–	10	3 (30.0)	
Status epilepticus			
None*	253	25 (9.9)	0.60 0.46 0.001
Unprovoked	31	4 (12.9)	
Febrile	5	0	
Other	2	2 (100)	
Multiple seizures in 1 day			
No	187	16 (8.6)	0.12
Yes	103	15 (14.6)	
Neonatal seizures†			
No	284	29 (10.2)	0.05
Yes	4	2 (50.0)	
Febrile seizures†			
No	247	29 (11.7)	0.15
Yes	41	2 (4.9)	

* Indicates reference group. For EEG, reference is to all others without specific characteristic.

† History unavailable for two.

Table 5 Associations between intractability and continuous factors in the children with nonidiopathic localization-related syndromes (*n* = 290)

Prognostic factor	Not intractable (<i>n</i> = 259), mean (SD)	Intractable (<i>n</i> = 31), mean (SD)	<i>p</i> Value
Age at onset, y	6.0 (4.0)	5.7 (4.6)	0.70
Log seizure frequency	0.9 (2.0)	2.4 (1.9)	0.0001
Log duration of epilepsy, mo	1.1 (1.5)	−0.1 (1.8)	0.0001
Log number of seizures	2.0 (1.8)	2.3 (1.9)	0.37

with localization-related epilepsy. Although previous studies have found an association, they did not distinguish status that occurred during the initial manifestations of epilepsy—before diagnosis and treatment—from status that occurred after diagnosis and treatment.^{4,12}

The nonsignificant, slightly protective association between febrile seizures and intractability is consistent with other studies that have not found febrile seizures to be associated with seizure outcome in childhood epilepsy.^{8,12} Previously, we found no association between febrile seizures and partial epilepsy, partial seizures, temporal lobe epilepsy, and complex partial seizures specifically.³⁶ This suggests that the association between febrile seizures and intractable epilepsy seen in surgical series is not a simple cause–effect relationship. Prolonged follow-up is needed to determine whether a second peak of intractability, perhaps associated with febrile seizures or with other factors, occurs later in life.

Although the risk of intractability was highest in cryptogenic/symptomatic generalized syndromes, half the intractability occurred in nonidiopathic localization-related syndromes. Initial seizure frequency and focal EEG slowing were powerful predictors of intractability in these children. This suggests

that, in children who present with nonidiopathic localization-related epilepsy, those with a few seizures scattered over a year are unlikely to become intractable, whereas those with the same number of seizures over a week or month are at substantially greater risk. This should not necessarily lead one to take extraordinary measures in children initially presenting with a high seizure frequency; however, such children might be observed and followed more closely.

The lack of a standard, accepted definition is a challenge in studying intractability.^{37,38} Our study was designed to identify children who developed drug-resistant epilepsy early in their seizure disorder and who therefore might be good candidates for more aggressive and new therapies, possibly even early surgical intervention. This requires intractability to be defined separately from remission, because failure to enter remission may be attributable to factors that preclude the designation of intractability. Such factors include not being treated, noncompliance, and relapse after recent discontinuation of medication.^{8,9} In fact, 10% of patients in our study and 20% in another recent study were untreated during the first year after diagnosis.^{10,11} We did not consider patients intractable if they were not in remission, for these and other similar reasons. There is no single definition of intractability, and the topic is complex.³⁸ Our definition addresses three fundamental considerations: medication failure, seizure frequency, and time. Other investigators have offered definitions of intractability,^{12–14,39,40} and all have considered these same factors in their definitions with highly comparable if not identical results.

Although some children with intractable epilepsy may eventually enjoy complete seizure control after many years,³⁹ avoiding a long period of uncontrolled seizures would be preferable if other options are available. In the past several years, at least eight new drugs have been approved in the United States; others are currently available elsewhere or are in development. Vagal nerve stimulators have been approved, the ketogenic diet is a recognized treatment, and the indications for epilepsy surgery have increased. Experience with many of these treatments in children is still limited. Early identification would permit selective targeting of those at greatest risk and in whom the benefits would be greatest for early aggressive therapy first in randomized trials and

Table 6 Multivariable model for predicting intractability in the group with nonidiopathic localization-related syndromes (*n* = 290)

Prognostic factor	Rate ratio	95% Confidence interval	<i>p</i> Value
Log seizure frequency	1.35	1.14, 1.59	0.001
Focal slowing	2.68	1.16, 6.16	0.02
“Other” status epilepticus	6.66	1.25, 35.41	0.03
Neonatal seizures	8.24	1.89, 35.99	0.005
Age 5–9*	0.43	0.16, 1.15	0.09
Unprovoked status*	1.81	0.62, 5.29	0.28
Symptomatic etiology*	1.09	0.46, 2.58	0.84
Febrile seizures*	0.53	0.13, 2.25	0.39
Log number of seizures*	0.83	0.63, 1.10	0.19
>1 seizure in a day*	0.64	0.23, 1.76	0.39
Age <1 y at onset*	2.01	0.76, 5.29	0.16

* Estimate is after adjustment for first four factors. The first four factors are not adjusted for nonsignificant terms.

later in routine clinical practice. Our data suggest that children at substantially greater than average risk of developing intractable epilepsy can be identified early.

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